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Integrated Circuitry, Array of Cross-Point Memory Cells, Method Used in Forming Integrated Circuitry

Abstract

Integrated circuitry comprises a horizontally-elongated insulative wall directly above a conductive node. The wall comprises insulative material. A conductive via extends through the wall to the conductive node. A conductive line is directly above the wall and directly above the conductive via. The conductive via directly electrically couples together the conductive line with the conductive node. Insulator material is longitudinally-along laterally-opposing sides of the wall. An interface of the insulative material of the wall and the insulator material are on each of the laterally-opposing sides of the wall. Other embodiments, including method, are disclosed.

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Background/Summary

RELATED PATENT DATA [0001] This patent resulted in from a divisional application of U.S. patent application Ser. No. 17/742,154, filed May 11, 2022, which is hereby incorporated by reference herein.

TECHNICAL FIELD

[0002] Embodiments disclosed herein pertain to integrated circuitry, to arrays of cross-point memory cells, and to methods used in forming integrated circuitry.

BACKGROUND

[0003] Memory is one type of integrated circuitry and is used in computer systems for storing data. Memory may be fabricated in one or more arrays of individual memory cells. Memory cells may be written to, or read from, using digitlines (which may also be referred to as bitlines, data lines, or sense lines) and access lines (which may also be referred to as wordlines). The sense lines may conductively interconnect memory cells along columns of the array, and the access lines may conductively interconnect memory cells along rows of the array. Each memory cell may be uniquely addressed through the combination of a sense line and an access line.

[0004] Memory cells may be volatile, semi-volatile, or non-volatile. Non-volatile memory cells can store data for extended periods of time in the absence of power. Non-volatile memory is conventionally specified to be memory having a retention time of at least about 10 years. Volatile memory dissipates and is therefore refreshed/rewritten to maintain data storage. Volatile memory may have a retention time of milliseconds or less. Regardless, memory cells are configured to retain or store memory in at least two different selectable states. In a binary system, the states are considered as either a “0” or a “1”. In other systems, at least some individual memory cells may be configured to store more than two levels or states of information.

[0005] The smallest and simplest memory cell will likely be comprised of two electrodes having a programmable material, and possibly a select device (such as a diode or ovonic threshold switch), received between them. Suitable programmable materials have two or more selectable memory states to enable storing of information by an individual memory cell. The reading of the cell determines which of the states the programmable material is in, and the writing of information to the cell places the programmable material in a predetermined state. Some programmable materials retain a memory state in the absence of refresh, and thus may be incorporated into nonvolatile memory cells.

[0006] Arrays of memory cells may comprise a plurality of access lines at one elevation and a plurality of sense lines at another elevation, with at least programmable material there-between. Individual memory cells are written to or read from by application of suitable voltage and/or current to the respective crossing access line and sense line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a diagrammatic cross-sectional view of a portion of a substrate in process in accordance with an embodiment of the invention and is taken through line 1-1 in FIG. 2.

[0008] FIG. 2 is an enlarged diagrammatic cross-sectional view taken through line 2-2 in FIG. 1.

[0009] FIGS. 3-24 are diagrammatic sequential sectional, expanded, enlarged, and/or partial views of the construction of FIGS. 1 and 2, or portions thereof, in process in accordance with some

embodiments of the invention.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0010] Embodiments of the invention include methods used in forming integrated circuitry and integrated circuitry independent of method of manufacture. Aspects of the invention were motivated in fabricating an array of cross-point memory cells, although the invention and all claims are not so limited. Example embodiments of methods of forming integrated circuitry are initially described with reference to FIGS. 1-24.

[0011] Referring to FIGS. 1 and 2, an example substrate construction **10** comprising an array or array area **12** (e.g., a memory array) has been fabricated relative to a base substrate **11**. Base substrate **11** may comprise any of conductive/conductor/conducting, semiconductive/semiconductor/semiconducting, and insulative/insulator/insulating (i.e., electrically herein) materials. Various materials are above base substrate **11**. Materials may be aside, elevationally inward, or elevationally outward of the FIGS. 1 and 2—depicted materials. For example, other partially or wholly fabricated components of integrated circuitry may be provided somewhere above, about, or within base substrate **11**. Control and/or other peripheral circuitry for operating components within an array (e.g., a cross-point memory array) may also be fabricated and may or may not be wholly or partially within a memory array or sub-array. Further, multiple sub-arrays may also be fabricated and operated independently, in tandem, or otherwise relative one another. As used in this document, a “sub-array” may also be considered as an array.

[0012] Example insulating material **14** has been formed directly above base substrate **11**, for example comprising doped and/or undoped silicon dioxide. Example laterally-spaced conductive nodes **16** (“laterally-spaced” in some vertical cross-section, for example that shown by FIG. 2) are in insulating material **14**, for example such being shown as conductive vias **16** extending therethrough to circuitry therebelow (not shown). The discussion largely proceeds with reference to two of laterally-spaced conductive nodes **16** as are shown in FIG. 2 and processing associated therewith. Such nodes need not be horizontally-aligned with one another as shown in FIG. 1 and, if not, need not horizontally overlap. Example conductive nodes **16** individually comprise a top **30**. In one embodiment, an example insulating material **15** (e.g., silicon nitride) has been formed above insulating material **14** and conductive nodes **16**.

[0013] Referring to FIGS. 3 and 4, first insulative material **14** (e.g., doped and/or undoped silicon dioxide, silicon nitride, or aluminum oxide) has been formed directly above two laterally-spaced conductive nodes **16**. A horizontally-elongated trench **18** has been formed (e.g., by anisotropic etching) in first insulative material **14**, with such being directly above and spanning laterally-between two laterally-spaced conductive nodes **16**. Multiple trenches **18** are shown, with the discussion largely proceeding with respect to the trench **18** as seen in the cross-section that is FIG. 4. In the depicted example, trench **18** has been formed to stop atop or within insulating material **15** and may be considered as having laterally-opposing sides **24** (e.g., in the vertical cross-section that is FIG. 4).

[0014] Referring to FIGS. 5 and 6, two horizontally-elongated and laterally-spaced walls **22** have been formed aside laterally-opposing sides **24** of horizontally-elongated trench **18**. Walls **22** comprise insulative material **20** (e.g., doped and/or undoped silicon dioxide, silicon nitride, or aluminum oxide). Individual walls **22** are directly above a respective one of two laterally-spaced conductive nodes **16**. An example way of forming walls **22** is by a maskless manner where a thin layer of second insulative material **20** (e.g., of a sub-lithographic thickness) is deposited atop first insulative material **14**, along trench sides **24**, and across the bottom of trench **18**, followed by maskless anisotropic etching thereof to substantially remove such from being over horizontal surfaces, resulting in the example wall structures **22** as shown.

[0015] In one embodiment, walls **22** may be considered as individually having longitudinally-opposing longitudinal ends **26**, **27**, with walls **22** being connected at their respective such longitudinal ends **26**, **27**. In one embodiment, first insulative material **14** and second insulative

material **20** are of the same composition relative one another, and in another embodiment are of different compositions relative one another. Individual walls **22** may be considered as comprising laterally-opposing sides **50**, a bottom **28**, and a top **52**. In one embodiment and as shown, individual walls **22** have bottom **28** vertically-spaced above top **30** of the respective individual laterally-spaced conductive node **16** directly there-under (e.g., by insulating material **15**). [0016] Referring to FIGS. **7** and **8**, space **32** that is laterally-between walls **22** has been filled with insulating material **36** (e.g., doped and/or undoped silicon dioxide, silicon nitride, or aluminum oxide), for example by depositing insulating material **36** to overfill space **32** and planarizing such back at least to top surfaces of first insulative material **14** and second insulative material **20**. Insulating material **36** and first insulative material **14** may be of the same composition or of different compositions relative one another. Ideally, second insulative material **20** is of different composition from each of first insulative material **14** and insulating material **36** such that a maskless selective etching thereof may occur with respect to materials **14** and **36** (as described below) to avoid a separate masking step.

[0017] Referring to FIGS. **9** and **10**, walls **22** have been vertically recessed relative to uppermost surfaces of first insulative material **14** and insulating material **36** (e.g., by etching material **20** selectively relative to materials **14** and **36**), for example thereby forming a horizontally-elongated trough **34** therein that is directly above individual walls **22**.

[0018] Referring to FIGS. **11** and **12**, and in one embodiment, first insulative material **14** and insulating material **36** have been etched (e.g., wet isotropic etching) to laterally-widen trough **34** to be laterally wider than individual walls **22**. In one such embodiment and as shown, such etching forms a bottom **40** of horizontally-elongated trough **34** to be lower than tops **52** of individual walls **22** (e.g., on both of laterally-opposing sides **50** of individual walls **22**).

[0019] Referring to FIGS. **13-15**, a contact opening **38** has been formed through individual walls **22** (e.g., by masked anisotropic etching of second insulative material **20**). Contact opening **38** extends to the respective one of two laterally-spaced conductive nodes **16** directly above which the respective individual wall **22** was formed.

[0020] Referring to FIGS. **16-19**, conductive material **42** has been formed in contact openings **38** and directly above vertically-recessed walls **22** (e.g., in troughs **34**). Conductive material **42** comprises a conductive via **44** in contact opening **38** and a conductive line **46** directly above individual vertically-recessed walls **22**. Conductive via **44** directly electrically couples together conductive line **46** with the respective one of two laterally-spaced conductive nodes **16** directly above which the respective individual wall **22** was formed. In one embodiment, the forming of conductive material **42** comprises forming a first conducting material **43** (e.g., titanium nitride) to line and less-than-fill horizontally-elongated trough **34** and the respective contact opening **38** directly therebelow. Then, a second conducting material **45** that is of different composition from that of first conducting material **43** (e.g., of higher conductivity than that of first conducting material **43**; e.g., elemental tungsten) is formed to fill remaining volumes of horizontally-elongated trough **34** and the respective contact opening **38** directly therebelow. In one such embodiment, the forming of second conducting material **45** is conducted selectively onto first conducting material **43** relative to exposed surfaces of first insulative material **14** and insulating material **36**. In another embodiment, the forming of second conducting material **45** is conducted non-selectively onto first conducting material **43** and onto exposed surfaces of first insulative material **14** and insulating material **36**. Regardless, conductive material **42** may ultimately be planarized back at least to top surfaces of first insulative material **14** and insulating material **36** to form the example depicted construction.

[0021] FIG. **16** shows, in one embodiment, conductive line **46** that is directly above one of two laterally-spaced conductive nodes **16** and conductive line **46** that is directly above the other of two laterally-spaced conductive nodes **16** are connected at their respective longitudinally-opposing longitudinal ends **48**, **49**. In one such embodiment, and referring to FIG. **20**, the method includes

severing (e.g., by masked anisotropic etching) of such conductive lines **46** at their respective longitudinally-opposing longitudinal ends **48, 49** so that said conductive lines **46** are not connected in a finished circuitry construction. In one embodiment, walls **22** are not so severed at their respective longitudinally-opposing longitudinal ends **26, 27** so that such walls **22** remain connected in the finished circuitry construction (e.g., FIG. **17**). Regardless, longitudinal ends **26, 27, 48,** and/or **49** may be within outside of array area **12**.

[0022] Other conductive vias (e.g., larger and not shown) may be formed within or outside of array area **12** commensurately with forming of conductive vias **44**.

[0023] In one embodiment, the integrated circuitry being fabricated comprises cross-point memory circuitry, for example as shown in FIGS. **21-24**. Such shows example subsequent processing whereby construction **10** comprises an array **12** of cross-point memory cells (e.g., **75**) that individually comprise part of a lower conducting line (e.g., the lower conducting line being one of lower conductive lines **46**), part of an upper conducting line (e.g., **70**) that crosses the lower conducting line, and programmable material (e.g., **72**) that is vertically-between the lower conducting line and the upper conducting line where such cross. A select device (not shown) may be associated with or comprise part of the individual memory cells. Example insulative material **80** is shown surrounding components **70** and **72** (e.g., doped and/or undoped silicon dioxide, silicon nitride, or aluminum oxide).

[0024] Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used with respect to the above-described embodiments.

[0025] In one embodiment, a method used in forming integrated circuitry comprises forming two horizontally-elongated and laterally-spaced insulative-material walls (e.g., **22**; e.g., regardless of forming a trench **18** and otherwise regardless of how such are formed). Individual of the walls are formed directly above a respective one of two laterally-spaced conductive nodes (e.g., **16**). The walls are vertically recessed relative to uppermost surfaces of insulator material (e.g., materials **14** and **36**) that is laterally-aside and laterally-between the individual walls. A contact opening (e.g., **38**) is formed through the individual walls. The contact opening extends to the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed. Conductive material (e.g., **42**) is formed in the contact openings and directly above the vertically-recessed walls. The conductive material comprises a conductive via (e.g., **44**) in the contact opening and a conductive line (e.g., **46**) directly above individual of the vertically-recessed walls. The conductive via directly electrically couples together the conductive line with the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed. In one embodiment, the insulative material and the insulator material are of the same composition relative one another and in another embodiment are not. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0026] Pitch multiplication may be used in fabrication of any of the above components. Combining the forming of the conductive material of conductive lines and conductive vias directly thereunder as described herein may improve or eliminate overlay margin issues.

[0027] Alternate embodiment constructions may result from method embodiments described above, or otherwise. Regardless, embodiments of the invention encompass memory arrays independent of method of manufacture. Nevertheless, such memory arrays may have any of the attributes as described herein in method embodiments. Likewise, the above-described method embodiments may incorporate, form, and/or have any of the attributes described with respect to device embodiments.

[0028] In one embodiment, integrated circuitry (e.g., comprising construction **10** in FIGS. **16-20** or construction **10** in FIGS. **21-24**) comprises a horizontally-elongated insulative wall (e.g., **22**) directly above a conductive node (e.g., **16**). The wall comprises insulative material (e.g., **20**). A conductive via (e.g., **44**) extends through the wall to the conductive node. A conductive line (e.g.,

46) is directly above the wall and directly above the conductive via. The conductive via directly electrically couples together the conductive line with the conductive node. Insulator material (e.g., materials **14** and **36**) is longitudinally-along laterally-opposing sides (e.g., **50**) of the wall. An interface (e.g., **55** in FIG. **19**) of the insulative material of the wall and the insulator material is on each of the laterally-opposing sides of the wall. In one embodiment, the interface is continuous all horizontally-along the wall and in another embodiment the interface is discontinuous at least somewhere horizontally-along the wall. The interface will be so continuous when the insulator material and the insulative material are of different compositions relative one another. The interface may or may not be so continuous when the insulator material and the insulative material are of the same composition relative to one another. For example, separate-in-time formed insulator material and insulative material of the same composition relative one another may nevertheless have a perceptible interface in a finished construction. Some of that interface may effectively disappear (i.e., not be perceptible) and some may remain perceptible whereby that interface is discontinuous in one or more locations longitudinally-there-along (e.g., as may occur by welding of the same-composition materials together due to subsequent heating during manufacture).

[0029] In one embodiment, the conductive line has a bottom (e.g., **60**) on each of the laterally-opposing sides of the wall that is below a top (e.g., **52**) of the wall. In one embodiment, the conductive line is laterally wider than the wall and in one such embodiment the conductive line extends laterally-outward beyond both of the laterally-opposing sides of the wall longitudinally along the wall. In one embodiment, the conductive line has a bottom that is below a top of the wall and in one such embodiment the conductive line is directly against at least one of the laterally-opposing sides of the wall, and in one such latter embodiment the conductive line is directly against both of the laterally-opposing sides of the wall.

[0030] In one embodiment, the insulative material and the insulator material are of the same composition relative one another and in another embodiment are of different compositions relative one another.

[0031] Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0032] In one embodiment, an array (e.g., **12**) of cross-point memory cells (e.g., **75**) comprises laterally-spaced lower conducting lines (e.g., **46**), laterally-spaced upper conducting lines (e.g., **70**) that cross the laterally-spaced lower conducting lines, and programmable material (e.g., **72**) of individual cross-point memory cells (e.g., **75**) vertically-between the laterally-spaced lower conducting lines and the laterally-spaced upper conducting lines where such cross. A horizontally-elongated insulative wall (e.g., **22**) is directly below individual of the lower conducting lines and is directly above a conductive node (e.g., **16**). The wall comprises insulative material (e.g., **20**). A conductive via (e.g., **44**) extends through the wall to the conductive node and directly electrically couples one of the lower conducting lines to the conductive node. Insulator material (e.g., **14** and **36**) is longitudinally-along laterally-opposing sides (e.g., **50**) of the wall. An interface (e.g., **55**) of the insulative material of the wall and the insulator material is on each of the laterally-opposing sides of the wall.

[0033] In one embodiment, the lower conducting lines are laterally wider than the walls, and in one such embodiment the lower conducting lines individually extend laterally-outward beyond both of the laterally-opposing sides of the respective wall therebelow longitudinally along the respective wall therebelow.

[0034] In one embodiment, the lower conducting lines individually have a bottom that is below a top of the respective wall therebelow, and in one such embodiment the lower conducting lines individually are directly against at least one of the laterally-opposing sides of the respective wall therebelow, and in one such latter embodiment, the conductive line is directly against both of the laterally-opposing sides of the wall.

[0035] In one embodiment, the lower conducting lines individually have a bottom on each of the

laterally-opposing sides of the respective wall therebelow that is below a top of the respective wall therebelow. In one embodiment, the insulative material and the insulator material are of the same composition relative one another and in another embodiment they are not.

[0036] Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0037] The above processing(s) or construction(s) may be considered as being relative to an array of components formed as or within a single stack or single deck of such components above or as part of an underlying base substrate (albeit, the single stack/deck may have multiple tiers). Control and/or other peripheral circuitry for operating or accessing such components within an array may also be formed anywhere as part of the finished construction, and in some embodiments may be under the array (e.g., CMOS under-array). Regardless, one or more additional such stack(s)/deck(s) may be provided or fabricated above and/or below that shown in the figures or described above. Further, the array(s) of components may be the same or different relative one another in different stacks/decks and different stacks/decks may be of the same thickness or of different thicknesses relative one another. Intervening structure may be provided between immediately-vertically-adjacent stacks/decks (e.g., additional circuitry and/or dielectric layers). Also, different stacks/decks may be electrically coupled relative one another. The multiple stacks/decks may be fabricated separately and sequentially (e.g., one atop another), or two or more stacks/decks may be fabricated at essentially the same time.

[0038] The assemblies and structures discussed above may be used in integrated circuits/circuitry and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multichip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

[0039] In this document unless otherwise indicated, “elevational”, “higher”, “upper”, “lower”, “top”, “atop”, “bottom”, “above”, “below”, “under”, “beneath”, “up”, and “down” are generally with reference to the vertical direction. “Horizontal” refers to a general direction (i.e., within 10 degrees) along a primary substrate surface and may be relative to which the substrate is processed during fabrication, and vertical is a direction generally orthogonal thereto. Reference to “exactly horizontal” is the direction along the primary substrate surface (i.e., no degrees there-from) and may be relative to which the substrate is processed during fabrication. Further, “vertical” and “horizontal” as used herein are generally perpendicular directions relative one another and independent of orientation of the substrate in three-dimensional space. Additionally, “elevationally-extending” and “extend (ing) elevationally” refer to a direction that is angled away by at least 45° from exactly horizontal. Further, “extend (ing) elevationally”, “elevationally-extending”, “extend (ing) horizontally”, “horizontally-extending” and the like with respect to a field effect transistor are with reference to orientation of the transistor's channel length along which current flows in operation between the source/drain regions. For bipolar junction transistors, “extend (ing) elevationally”, “elevationally-extending”, “extend (ing) horizontally”, “horizontally-extending” and the like, are with reference to orientation of the base length along which current flows in operation between the emitter and collector. In some embodiments, any component, feature, and/or region that extends elevationally extends vertically or within 10° of vertical.

[0040] Further, “directly above”, “directly below”, and “directly under” require at least some lateral overlap (i.e., horizontally) of two stated regions/materials/components relative one another. Also, use of “above” not preceded by “directly” only requires that some portion of the stated region/material/component that is above the other be elevationally outward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components). Analogously, use of “below” and “under” not preceded by

“directly” only requires that some portion of the stated region/material/component that is below/under the other be elevationally inward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components).

[0041] Any of the materials, regions, and structures described herein may be homogenous or non-homogenous, and regardless may be continuous or discontinuous over any material which such overlies. Where one or more example composition(s) is/are provided for any material, that material may comprise, consist essentially of, or consist of such one or more composition(s). Further, unless otherwise stated, each material may be formed using any suitable existing or future-developed technique, with atomic layer deposition, chemical vapor deposition, physical vapor deposition, epitaxial growth, diffusion doping, and ion implanting being examples.

[0042] Additionally, “thickness” by itself (no preceding directional adjective) is defined as the mean straight-line distance through a given material or region perpendicularly from a closest surface of an immediately-adjacent material of different composition or of an immediately-adjacent region. Additionally, the various materials or regions described herein may be of substantially constant thickness or of variable thicknesses. If of variable thickness, thickness refers to average thickness unless otherwise indicated, and such material or region will have some minimum thickness and some maximum thickness due to the thickness being variable. As used herein, “different composition” only requires those portions of two stated materials or regions that may be directly against one another to be chemically and/or physically different, for example if such materials or regions are not homogenous. If the two stated materials or regions are not directly against one another, “different composition” only requires that those portions of the two stated materials or regions that are closest to one another be chemically and/or physically different if such materials or regions are not homogenous. In this document, a material, region, or structure is “directly against” another when there is at least some physical touching contact of the stated materials, regions, or structures relative one another. In contrast, “over”, “on”, “adjacent”, “along”, and “against” not preceded by “directly” encompass “directly against” as well as construction where intervening material(s), region(s), or structure(s) result(s) in no physical touching contact of the stated materials, regions, or structures relative one another.

[0043] Herein, regions-materials-components are “electrically coupled” relative one another if in normal operation electric current is capable of continuously flowing from one to the other and does so predominately by movement of subatomic positive and/or negative charges when such are sufficiently generated. Another electronic component may be between and electrically coupled to the regions-materials-components. In contrast, when regions-materials-components are referred to as being “directly electrically coupled”, no intervening electronic component (e.g., no diode, transistor, resistor, transducer, switch, fuse, etc.) is between the directly electrically coupled regions-materials-components.

[0044] Any use of “row” and “column” in this document is for convenience in distinguishing one series or orientation of features from another series or orientation of features and along which components have been or may be formed. “Row” and “column” are used synonymously with respect to any series of regions, components, and/or features independent of function. Regardless, the rows may be straight and/or curved and/or parallel and/or not parallel relative one another, as may be the columns. Further, the rows and columns may intersect relative one another at 90° or at one or more other angles (i.e., other than the straight angle).

[0045] The composition of any of the conductive/conductor/conducting materials herein may be conductive metal material and/or conductively-doped semiconductive/semiconductor/semiconducting material. “Metal material” is any one or combination of an elemental metal, any mixture or alloy of two or more elemental metals, and any one or more metallic compound(s).

[0046] Herein, any use of “selective” as to etch, etching, removing, removal, depositing, forming, and/or formation is such an act of one stated material relative to another stated material(s) so acted

upon at a rate of at least 2:1 by volume. Further, any use of selectively depositing, selectively growing, or selectively forming is depositing, growing, or forming one material relative to another stated material or materials at a rate of at least 2:1 by volume for at least the first 75 Angstroms of depositing, growing, or forming.

[0047] Unless otherwise indicated, use of “or” herein encompasses either and both.

CONCLUSION

[0048] In some embodiments, a method used in forming integrated circuitry comprises forming two horizontally-elongated and laterally-spaced insulative-material walls. Individual of the walls are formed directly above a respective one of two laterally-spaced conductive nodes. The walls are vertically recessed relative to uppermost surfaces of insulator material that is laterally-aside and laterally-between the individual walls. A contact opening is formed through the individual walls. The contact opening extends to the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed. Conductive material is formed in the contact openings and directly above the vertically-recessed walls. The conductive material comprises a conductive via in the contact opening and a conductive line directly above individual of the vertically-recessed walls. The conductive via directly electrically couples together the conductive line with the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed.

[0049] In some embodiments, a method used in forming integrated circuitry comprises forming first insulative material directly above two laterally-spaced conductive nodes. A horizontally-elongated trench is formed in the first insulative material. The horizontally-elongated trench is directly above and spans laterally-between the two laterally-spaced conductive nodes. Two horizontally-elongated laterally-spaced second insulative material walls are formed aside laterally-opposing sides of the horizontally-elongated trench. Individual of the walls are directly above a respective one of the two laterally-spaced conductive nodes. Space that is laterally-between the walls is filled with insulating material. The walls are vertically recessed relative to uppermost surfaces of the first insulative material and the insulating material. A contact opening is formed through the individual walls. The contact opening extends to the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed. Conductive material is formed in the contact openings and directly above the vertically-recessed walls. The conductive material comprises a conductive via in the contact opening and a conductive line directly above individual of the vertically-recessed walls. The conductive via directly electrically couples together the conductive line with the respective one of the two laterally-spaced conductive nodes directly above which the respective individual wall was formed.

[0050] In some embodiments, integrated circuitry comprises a horizontally-elongated insulative wall directly above a conductive node. The wall comprises insulative material. A conductive via extends through the wall to the conductive node. A conductive line is directly above the wall and directly above the conductive via. The conductive via directly electrically couples together the conductive line with the conductive node. Insulator material is longitudinally-along laterally-opposing sides of the wall. An interface of the insulative material of the wall and the insulator material are on each of the laterally-opposing sides of the wall.

[0051] In some embodiments, an array of cross-point memory cells comprises laterally-spaced lower conducting lines, laterally-spaced upper conducting lines that cross the laterally-spaced lower conducting lines, and programmable material of individual cross-point memory cells vertically-between the laterally-spaced lower conducting lines and the laterally-spaced upper conducting lines where such cross. A horizontally-elongated insulative wall is directly below individual of the lower conducting lines and that is directly above a conductive node. The wall comprises insulative material. A conductive via extends through the wall to the conductive node and that directly electrically couples one of the lower conducting lines to the conductive node. Insulator material is longitudinally-along laterally-opposing sides of the wall. An interface of the insulative material of

the wall and the insulator material is on each of the laterally-opposing sides of the wall.
[0052] In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

Claims

1. Integrated circuitry comprising: a horizontally-elongated insulative wall directly above a conductive node, the wall comprising insulative material; a conductive via extending through the wall to the conductive node; a conductive line directly above the wall and directly above the conductive via, the conductive via directly electrically coupling together the conductive line with the conductive node; and insulator material longitudinally-along laterally-opposing sides of the wall, an interface of the insulative material of the wall and the insulator material on each of the laterally-opposing sides of the wall.
2. The integrated circuitry of claim 1 wherein the conductive line is laterally wider than the wall.
3. The integrated circuitry of claim 2 wherein the conductive line extends laterally-outward beyond both of the laterally-opposing sides of the wall longitudinally along the wall.
4. The integrated circuitry of claim 1 wherein the conductive line has a bottom that is below a top of the wall.
5. The integrated circuitry of claim 4 wherein the conductive line is directly against at least one of the laterally-opposing sides of the wall.
6. The integrated circuitry of claim 5 wherein the conductive line is directly against both of the laterally-opposing sides of the wall.
7. The integrated circuitry of claim 1 wherein the conductive line has a bottom on each of the laterally-opposing sides of the wall that is below a top of the wall.
8. The integrated circuitry of claim 1 wherein the interface is continuous all horizontally-along the wall.
9. The integrated circuitry of claim 1 wherein the interface is discontinuous at least somewhere horizontally-along the wall.
10. The integrated circuitry of claim 1 wherein the insulative material and the insulator material are of the same composition relative one another.
11. The integrated circuitry of claim 1 wherein the insulative material and the insulator material are of different compositions relative one another.
12. An array of cross-point memory cells, comprising: laterally-spaced lower conducting lines, laterally-spaced upper conducting lines that cross the laterally-spaced lower conducting lines, and programmable material of individual cross-point memory cells vertically-between the laterally-spaced lower conducting lines and the laterally-spaced upper conducting lines where such cross; a horizontally-elongated insulative wall directly below individual of the lower conducting lines and that is directly above a conductive node, the wall comprising insulative material; a conductive via extending through the wall to the conductive node and that directly electrically couples one of the lower conducting lines to the conductive node; and insulator material longitudinally-along laterally-opposing sides of the wall, an interface of the insulative material of the wall and the insulator material on each of the laterally-opposing sides of the wall.
13. The array of claim 12 wherein the lower conducting lines are laterally wider than the walls.
14. The array of claim 13 wherein the lower conducting lines individually extend laterally-outward beyond both of the laterally-opposing sides of the respective wall therebelow longitudinally along the respective wall therebelow.

- 15.** The array of claim 12 wherein the lower conducting lines individually have a bottom that is below a top of the respective wall therebelow.
- 16.** The array of claim 15 wherein the lower conducting lines individually are directly against at least one of the laterally-opposing sides of the respective wall therebelow.
- 17.** The integrated circuitry of claim 16 wherein the conductive line is directly against both of the laterally-opposing sides of the wall.
- 18.** The array of claim 12 wherein the lower conducting lines individually have a bottom on each of the laterally-opposing sides of the respective wall therebelow that is below a top of the respective wall therebelow.
- 19.** The array of claim 12 wherein the interface is continuous all horizontally-along the wall.
- 20.** The array of claim 12 wherein the interface is discontinuous at least somewhere horizontally-along the wall.
- 21.** The array of claim 12 wherein the insulative material and the insulator material are of the same composition relative one another.
- 22.** The array of claim 12 wherein the insulative material and the insulator material are of different compositions relative one another.
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